MILITARY SPECIFICATION

WELDING MACHINE, ARC, DIRECT CURRENT (DC), CONSTANT CURRENT (CC), CONSTANT VOLTAGE (CV), 350 AMPERE , DIESEL ENGINE DRIVEN (DED) , BASE MOUNTED

This specification is approved for use by the US Army Armament, Research and Development Engineering Center (Engineering Support Directorate, Tool and Equipment Engineering Division, Standardization and Specification Engineering Branch), Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers a base mounted, DC, cc and CV, DED welding machine of 350 ampere capacity.

2 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

FEDERAL

W-R-550 - Rod, Ground (With Attachment)
QQ-S-781 - Strapping, Steel, and Seals
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-P-40 - Packaging and Packing of Hand Tools

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN : SMCAR-EST-S, Rock Island, IL 61299-7300 by using a self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-C-104 - Crates, Wood;Lumber and Plywood Sheated, Nailed and Bolted
MIL-P-116 - Preservation, Methods of
MIL-V-173 - Varnish, Moisture- and Fungus-Resistant (For Treatment of Communications, Electronics and Associated Equipment)
MIL-P-514 - Plate, Identification, Instruction and Marking Blank
MIL-L-2104 - Lubricating Oil, Internal Combustion Engine, Tactical Service
MIL-C-3600 - Compressors, Rotory, Power Driven; and Compressors, Reciprocating, Power Driven: Air and Gas (Except Oxygen and Refrigerant), Packaging of
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts):Packaging of
MIL-T-45542 - Tool Sets, Shop Sets and Kits Maintenance, Modification and Tool Packaging of
MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging Of
MIL-L-46152 - Lubricating Oil, Internal Combustion Engine Administrative Service
MIL-A-46153 - Antibreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package
MIL-W-52161 - Welding Set, Inert Gas Shielded Arc, Consumable Metal Electrode

STANDARD

FEDERAL

FED-STD-H28 - Screw-Threads Standards For Federal Service

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-1186 - Cushioning, Anchoring Bracing, Blocking and Waterproofing; with Appropriate Test Methods

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

ARMY

11021251 - Finish Methods
11021420 - Cable Assy, Ground Clamp
11021430 - Cable Assy, Electrode Holder
11021513 - Panel Assy, Upper Control
11021554 - Panel Assy, Rear Lower
11021570 - Placard

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11021573 - Placard Assy, Connection Diag., Lube Instructions
11021744 - Safety Shutdown Installation Assembly
11021792 - Generator Assy
11022100 - Welding Machine, Arc, DC, CC-CV, Gen., DED, Base Mounted
11022184 - Switch Assembly, Mode Selector
11022512 - Nameplate, Rating
11022586 - Diode, Silicone , 275 Amp, Negative Base
11022587 - Diode, Silicone , 275 Amp, Positive Base
11022659 - Nameplate
11022704 - Air Compressor Kit Assy

(Copies of specifications, standards, handbooks, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURER'S ASSOCIATION

EW- 1 - Electric Arc Welding Standards

(Application for copies should be addressed to the National Electrical Manufacturer's Association, 2101 L Street NW, Washington, DC 20037.)

AMERICAN WELDING SOCIETY

A5.18 - Specification for Carbon Steel Filler Metals
for Gas Shielded Arc Welding

(Applications for copies should be addressed to the American Welding Society, 550 NW Le Jeune Road, P.O. Box 351040, Miami, FL 33135.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3 REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1 and 6.2.1).

3.2 Design. All parts of the welding machine and its components shall be new. The welding machine shall be in accordance with Drawing Number 11022100, and as specified herein. Welding terms and definitions used in this specification are in accordance with NEMA EW-1. When specified (see 6.2.1), the welding machine shall be furnished with the completely mounted air compressor kit assembly in accordance with Drawing Number 11022704.

3.2.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g., shop drawings, layouts, flow-sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2.2 Measurement system. Either the US customary (Inch) system of units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. when only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes and capacities are given in the Inch and SI system of units.

3.3 Construction. The welding machine and its components shall be free from any characteristics or defects which may render the machine unsuitable or inefficient for the purpose intended. The machine and its equipment shall be so constructed as to be capable of performing the operations specified herein.

3.3.1 Materials. Materials shall be as specified herein and as shown on drawing 11022100 and the applicable drawings referenced therein. Materials not specifically designated shall be selected by the contractor and shall be subject to all provisions of this specification. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

3.3.2 Castings and forgings. Castings shall be free of visible blowholes, porosity, hardspots, shrinkage defects, cracks, or other defects. Forgings shall be free of visible scale, inclusions, cold shuts, mismatching, or other defects. In no event shall such processes as welding, peening, plugging, or filling with cold solders or metallic pastes be used on castings or forgings.

3.3.3 Welding, brazing and soldering. Welding, brazing and soldering shall be employed only where those operations are included in fabrication of the original design. These operations shall not be employed as repair measures for defective parts.

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3.3.4 Fastening devices. All fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.3.5 Threads. All threaded parts of the welding machine and its components shall conform to FED-STD-H28 and applicable "Detailed Standard" section referenced therein.

3.3.6 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any part may be replaced, interchanged, and adjusted without modification of the replacement part or any parts of the unit. All replaceable parts and assemblies shall be permanently and legibly marked in accordance with MIL-STD-130.

3.3.7 Lubrication. The contractor shall include and install all lubricants, fluids, greases and other compounds ready for operation upon arrival to the Government. In the event a lubricant reservoir cannot br covered and sealed, the appropriate lubricant shall be provided in a sealed container and shipped with each welding machine. A warning tag shall be placed near the filling area, warning the user to install the furnished lubricant before operation of the equipment. The operator's manual shall also reflect the above warning.

3.3.8 Cleaning, treating and painting. Surfaces to be painted as specified on Drawing 11022100 and applicable drawings referenced therein shall be cleaned and dried to ensure they are free of contaminates such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion products or any other interfering substances. After cleaning and before any corrosion product or other coating interfering material can be deposited, the surface shall be prepared or treated to ensure adhesion of the coating system. Finish methods shall be as specified on the individual drawings and drawing 11021251. The total dry film thickness shall not be less than 2.5 roils over the entire surface. The color of the finish coat shall be lusterless forest green infrared absorbent paint. The primer shall be applied as specified on drawing number 11021251.

3.3.9 Fungus and moisture resistance. The electrical circuitry specified on Drawing 11022100 and applicable drawings referenced therein shall be protected from fungus growth and moisture by an overall fungus treatment in accordance with MIL-V-173 with the exception that:

a. Components of circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely effected by the coating shall not be treated.

3.3.10 Welding leads (drawing number 11021420 and 11021430). The machine shall be equipped with one 50 foot (15.24 meters) electrode lead and one 50 foot (15.24 meters) work lead.

3.3.11 Nameplates (drawing number 11022659 and 11022512). The nameplates shall contain the information listed below. The approved item name shall be: Welding Machine Arc, DC, CC-CV, Gen., DED, Base Mounted.

(a) Nomenclature,
(b) Manufacture's Name,
(c) Manufacturer's Model Number
(d) Manufacture's Serial Number,
(e) Engine Make, Model, and Serial Number
(f) Constant Current Characteristics (CC)

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(g) Constant Voltage Characteristics (CV)
(h) Auxiliary Power KVA, AC Voltage, Frequency, Phase,
(i) Date of Manufacturer
(j) Gross Weight
(k) Contract or Order Number
(l) National Stock Number.

The nameplate shall be marked with the above information in accordance with MIL-STD-130. The plate shall be aluminum conforming with Type III, Composition A, Class 2 of MIL-P-514. The plate shall be attached to the equipment as specified on Drawing Number 11021554.

3.3.12 Wiring diagram (drawing number 11022501). A wiring diagram depicting actual machine wiring of the welding machine, protected with a transparent covering or coating, shall be provided and fastened to the inside surface of the left door nearest the control panel, when facing the control panel.

3.3.13 Lubrication chart and engine wiring diagram (drawing number 11021573). A lubrication chart and engine wiring diagram protected with a transparent covering or coating shall be provided. The chart and diagram shall be permanently and securely attached to the inside surface of the left door nearest the control panel, when facing the control panel.

3.3.14 Ground rod. The welding machine shall be furnished with a Rod, Ground; steel, copper clad, 5/8 inch (in) (15.875 millimeters (mm)) diameter, three 3 foot (.914 meter (m)) sections with 6 foot (1.829 meters) cable No. 6 AWG with ground terminal and ground clamp, in accordance with Federal Specification W-R-550, Type III, Class B. Bus bar and hardware to mount ground terminal to the welder ground stud shall be included.

3.3.15 Selector switch (drawing number 11022184). The welding machine shall be furnished with a control panel mounted switch to provide for proper connection and operation for accommodating and utilizing the Welding Set, Arc, Inert Gas Shielded in accordance with Military Specification MIL-W-52161.

3.3.16 Ignition lock and keys. The welding machine shall be furnished with an ignition lock and two identical ignition keys. The ignition key and lock for each welding machine shall be different for a series of no less than 30 machines. The key series and lock shall repeat on each group of machines.

3.3.17 Lubricating oil.

3.3.17.1 Engine. Engine lubricating oil shall be in accordance with MIL-L-2104 or MIL-46152 per engine manufacturers recommendations.

3.3.17.2 Air compressor. The air compressor (drawing 11022704) lubricating oil shall be in accordance with MIL-L-46152 or MIL-L-2104 (A or B).

3.4 Performance and product characteristics.

3.4.1 Physical data.

3.4.1.1 Weight. The operational ready weight - including engine oil; engine coolant (55 percent ethylene glycol, 45 percent water by volume); full tank of fuel; wet batteries; etc. - shall not exceed 2,637 pounds (lbs) (1196 kilograms (kg)) with the air compressor kit and shall not exceed 2,405 lbs (1080 kg) without the air compressor kit.

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3.4.1.2 Dimensions. The welding machine overall dimensions shall be as follows:

Length	77-7/16 in. ± 3/8 in. 1966 mm ± 9mm	Width w/o compressor	31 in. 787 mm
Height over canopy	44-5/16 in. ± 3/8 in. 1125mm ± 9mm	Width w/o compressor	41-1/4 in. 1047 mm
Height over lifting eye	49-1/2 in. ± 3/8 in. 1257 mm ± 9mm		

3.4.2 Transportability. The welding machine, its cabinet or frame shall not deform in meeting the requirements in 3.4.2.1 and 3.4.2.2.

3.4.2.1 By lifting eye. The welding machine shall be equipped with a lifting eye as 'specified on the applicable drawing and the lifting eye shall be capable of supporting at least two and one-half times the operational ready weight as specified in 3.4.l.l.

3.4.2.2 By forklift. The welding machine shall be capable of sustaining elevated forklift transportability for a minimum distance of 50 feet (15 m) without damage.

3.4.3 Environmental conditions.

3.4.3.1 Low temperature. The welding machine shall be capable of being started and operating down to a minimum temperature of -25 degrees F (-31.67 degrees C). The welding machine shall not be adversely affected by storage down to a minimum temperature -28 degrees F (-33.4 degrees C) for at least 8 hours.

3.4.3.2 High temperature. The welding machine shall be capable of being started and operating up to a maximum temperature of +120 degrees F (+48.9 degrees C). The welding machine shall not be adversely affected by storage for at least 8 hours up to a maximum temperature of +160 degrees F (+71.1 degrees C). In addition the welding machine shall not be adversely affected by a relative humidity between 80 and 100 percent during high temperature operation and storage.

3.4.3.3 High altitude. The welding machine shall be capable of producing full rated output per NEMA EW-1 at 6,000 feet (1828 m) above sea level.

3.4.4 Engine safety devices.

3.4.4.1 LOW oil pressure shutdown device (drawing number 11021744). The engine shall be equipped with a lubricating oil low pressure shutdown device to stop the engine in the event of low oil pressure. The device shall stop the engine when the oil pressure drops to 10 psi (68.95 kilopascals (KPa)) or less.

3.4.4.2 High coolant temperature shutdown device (drawing number 11021744). The engine shall be equipped with a high cools.rl~ temperature shutdown device to stop the engine in the event of excessive coolant temperature. The device shall stop the engine when the coolant temperature exceeds +205 degrees F (+96.11 degrees C).

3.4.4.3 Overspeed shutdown device (drawing number 11021511). The engine shall be equipped with an overspeed shutdown device to stop the engine in the event of overspeed. The device shall stop the engine when the speed exceeds 2,200 revolutions per minute (rpm).

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3.4.5 Auxiliary power. The voltage regulation of the auxiliary power supply shall be plus/minus 3 percent at rated voltage (120V AC) from no load to full load (0 to 4 KVA).

3.4.6 Welding generator (drawing number 11021792). The welding generator shall conform to NEMA EW-1 Class I, 100 percent duty cycle, and the applicable drawings.

3.4.6.1 Welding output rating range. The welding output rating power range shall be in accordance with NEMA EW-1 for Class 1 (100 percent) DC Generator-Rectifier Arc welding machines. The controls shall permit welding by the short circuit transfer method without visible spatter.

3.4.6.2 Circuit voltage. At operating speed of 1,800 RPM the DC open circuit voltage at the welder output terminals shall not exceed 100 volts and the ripple voltage shall not exceed 10 percent. In the constant current mode (CC), the steady state short circuit current shall not be greater than 200 percent nor less than 110 percent of the normal current for that setting.

3.4.6.3 Welding performance. The welding machine, when equipped with the proper accessories, shall be capable of welding, utilizing the following processes:

SMAW (Shielded Metal Arc Welding)
GNAW (Gas Metal Arc Welding, including the Shorting Arc Process)
FCAW (Flux Cored Arc Welding)
GTAW (DC) (Gas Tungsten Arc Welding)
AAC (Air Arc Cutting - Carbon Arc Cutting)

All of these processes shall be within the current ranges as specified in NEMA EW-1 for the rating of the machine.

3.5 Rectifiers (11022586 and 11022587). The operating temperatures of the rectifiers under maximum load conditions shall not exceed a value which will damage the rectifiers or adversely affect the performance of the power supply.

3.6 Control panel (drawing number 11021513). All instruments and controls shall be legibly marked as to their function.

3.7 Dielectric strength. The insulation of all machine components (excluding rectifiers and electronic components) shall meet the requirements of NEMA EW-l, High Potential Tests.

3.8 Stability. The machine shall operate satisfactorily when inclined at an angle of 15 degrees in any direction from the horizontal plane.

3.9 Operating speed. The operating speed shall be 1,800 revolutions per minute (RPM). The variation in operating speed from a no load condition to the full rated load condition shall not exceed plus-or minus 3 percent.

3.10 Antifreeze. The engine cooling system shall be protected for a minimum of -40 degrees F (-40 degrees C) (55 percent ethylene glycol, 45 percent water by volume) with antifreeze conforming to MIL-A-46153.

3.11 Fuel tank. The fuel tank shall be drained of fuel oil for shipment after testing of the machine.

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3.12 Workmanship. Standards of workmanship shall be such as to assure equipment of a neat appearance, having the stability and efficient operating characteristics as found in commercial equipment of the type specified herein.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. When specified (see 6.2.1), first article inspection shall be performed on the welding machine. This inspection shall include the examinations and tests specified in Table I.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be applied to production units being offered for acceptance under the contract. The sampling for test and examination shall be in accordance with Table I. The contractor shall provide to the Government representative objective evidence that the examination requirements have been met. Failure of a welding machine to meet the sample inspection tests shall be cause for the Government to suspend acceptance until:

(a) The contractor has completed a failure analysis.
(b) The applicable production processes and procedures to prevent recurrence of the same defect in future production are evaluated and corrected.
(c) The welding machines awaiting submittal for Government acceptance, (including components and subassemblies at in-process or final assembly) are examined to ensure that the material containing the same defect(s) is purged from the inventory and not presented to the Government for acceptance.
(d) Objective evidence of the results of the failure analysis and corrective actions are provided to the Government representative prior to submitting the reconditioned welding machine for retest.

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TABLE I. Tests and examinations.

FIRST ARTICLE	QUALITY CONFORMANCE INSPECTION		TEST AND EXAMINATION	TEST PARA.	REQUIREMENT PARA.
	SAMPLING (1 OUT OF EA. 40)	100 PERCENT INSPECTION (EA. UNIT SHALL BE INSPECTED)			
X	-	-	Design (Drawing 11022100)	4.3	3.2
X	-	X	Construction	4.4	3.3
X	-	X	Material	4.4	3.3.1
X	-	X	Castings and forgings	4.4	3.3.2
X	-	X	Welding, Brazing and Soldering	4.4	3.3.3
X	-	X	Fastening devices	4.4	3.3.4
X	-	X	Threads	4.4	3.3 5
X	-	X	Interchangeability	4.4	3.3.6
X	-	X	Lubrication	4.4	3.3.7
X	-	X	Cleaning, treating and painting	4.4	3.3.8
X	-	X	Fungus and moisture resistance	4.4	3.3.9
X	-	X	Welding leads	4.4	3.3.10
X	-	X	Nameplate	4.4	3.3.11
X	-	X	Wiring Diagram	4.4	3.3.12
X	-	X	Lubrication chart and engine wiring diagram	4.4	3.3.13
X	-	X	Ground rod	4.4	3.3.14
X	-	X	Selector switch	4.4	3.3.15
X	-	X	Ignition keys	4.4	3.3.16
X	-	X	Engine oil	4.4	3.3.17.1
X	-	X	Compressor oil	4.4	3.3.17.2
X	-	-	Weight	4.5.1	3.4.1.1

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TABLE I. Tests and examinations-continued.

FIRST ARTICLE	QUALITY CONFORMANCE INSPECTION		TEST AND EXAMINATION	TEST PARA.	REQUIREMENT PARA.
	SAMPLING (1 OUT OF EA. 40)	100 PERCENT INSPECTION (EA. UNIT SHALL BE INSPECTED)			
X	X	-	Dimensions	4.5.2	3.4.1.2
X	-	-	Transport-ability (by lifting eye)	4.6.1	3.4.2.1
X	-	-	Transport-ability (by forklift)	4.6.2	3.4.2.2
-	-	-	Environmental conditions	4.7	3.4.3
X	-	-	Low Temperature	4.7.1	3.4.3.1
X	-	-	High temper-ature	4.7.2	3.4.3.2
X	-	-	High altitude	4.7.3	3.4.3.3
X	-	-	Low oil pressure shutdown device	4.8.1	3.4.4.1
X	-	-	High coolant temperature shutdown device	4.8.2	3.4.4.2
X	-	-	Overspeed shut-down device	4.8.3	3.4.4.3
X	-	-	Auxiliary power	4.9	3.4.5
-	-	-	Welding generator	4.10	3.4.6
X	-	-	Welding output rating range	4.10.1	3.4.6.1
X	X	-	Circuit voltage	4.10.2	3.4.6.2
X	X	-	Welding perfor-mance	4.10.3	3.4.6.3
X	-	-	Rectifiers	4.11	3.5
X	-	X	Control panel	4.12	3.6
X	X	-	Dielectric strength	4.13	3.7
X	-	-	Stability	4.14	3.8
X	X	-	Operating speed	4.15	3.9
X	-	X	Antifreeze	4.16	3.10
X	-	X	Fuel tank	4.17	3.11
X	-	X	Workmanship	4.18	3.12
X	-	X	Packaging inspection	4.19	5.

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4.3 Design. Visual examination shall be performed to ascertain compliance with Drawing 11022100.

4.4 Construction. The inspection of the welding machine shall include visual and manual examinations to determine conformance with the requirements of 3.3 through 3.3.17. Visual examination shall include verification of proper cleaning; presence of specified protective coatings; completeness of manufacture and assembly; adequacy of marking; repetition of key series; and correctness of engine and air compressor lubrication oil. Manual examinations shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at any practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the welding machine and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection.

4.5 Physical data.

4.5.1 Weight. The complete welding machine shall be weighed as per paragraph 3.4.1.1$_0$.

4.5.2 Dimensions. Using a steel measuring tape, the dimensions of the welding machine shall be measured to assure compliance with 3.4.1.2.

4.6 Transportability.

4.6.1 By lifting eye. To assure compliance with 3.4.2.1 attach an additional weight to the base of the welding machine to provide a total weight equal to 2-1/2 times the operational ready weight of the machine. Lift the unit two times by the machine lifting eye to a height of 4 feet (1.219 m). There shall be no visual evidence of permanent deformation, weld cracks, or other damage to the welding machine.

4.6.2 By forklift. To assure compliance with 3.4.2.2 place a chain through the lifting eye of the welding machine (no additional weight required) and over the tines of a forklift truck. Transport the unit in an elevated position for a distance of at least 50 feet.

4.7 Environmental conditions. A warmup time of at least 5 minutes shall be allowed before environmental testing begins (see 4.7.1, 4.7.2, and 4.7.3). Maximum power testing shall be performed for 5 continuous minutes.

4.7.1 Low temperature. To assure compliance with 3.4.3.1 the welding machine shall be submitted to storage at a minimum temperature of -28 degrees F (-33.4 degrees C) for a period of 24 hours. At the end of this period, the temperature shall rise to -25 to -20 degrees F (-31.67 to -28.89 degrees C) and after stabilization, the welding machine shall be started and the output of both the auxiliary power generator and the welding generator shall be checked for minimum and maximum output per NEMA EW-1 Class 1(100), using a resistance load bank having a power factor of 0.99 or higher.

4.7.2 High temperature. To assure compliance with 3.4.3.2 the welding machine shall be submitted to storage at a minimum temperature of +160 degrees F (71.1 degrees C) and induced relative humidity of 80 to 100 percent for a period of 24 hours. At the end of this period, the temperature shall be reduced to +120 to +115 degrees F (+48.9 to +46.12 degrees C) and 80 to 100 percent relative humidity. After a 24 hour

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stabilization period, the welding machine shall be started and the output of both the auxiliary power generator and the welding generator shall be checked for minimum and maximum output per NEMA EW-1 Class 1(100) using a resistance load bank having a power factor of 0.99 or higher.

4.7.3 High altitude. To assure compliance with 3.4.3.3 the welding machine shall be exposed to a condition of 6,000 feet (1828.8 m) above sea level or to a condition simulating an altitude of 6,000 feet (1828.8 m) above sea level (23.88 inches of mercury) (80.866 KPa). The unit shall be started and the output of both the auxiliary power generator and the welding generator shall be checked for minimum and maximum output per NEMA EW-1 Class 1(100), using a resistance load bank having a power factor of 0.99 or higher.

4.8 Engine safety devices.

4.8.1 Low oil pressure shutdown device (Drawing number 11021744). To assure compliance with 3.4.4.1 the low oil pressure shutdown device shall be tested with the engine running. Through the use of appropriate plumbing the oil pressure to the device shall be reduced to 10 psi (68.95 KPa) or less, without actually reducing the engine oil pressure. Failure of the device to shut down the engine shall be cause for rejection of the welder.

4.8.2 High coolant temperature shutdown device (Drawing number 11021744). To assure compliance with 3.4.4.2 the high coolant temperature shutdown device shall be tested with the engine running. The coolant temperature shall be elevated to a maximum of +220 degrees F (104.45 degrees C) by blocking the radiator or by any other appropriate means. Failure of the device to shutdown the engine when the coolant temperature exceeds +205 degrees F shall be cause for rejection of the welder.

4.8,3 Overspeed shutdown device (Drawing number 11021792). To assure compliance with 3.4.4.3 the overspeed safety device shall be tested with the engine running. The engine speed shall be manually adjusted to an overspeed setting of 2,200 rpm. Failure of the overspeed device to shutdown the engine shall be cause for rejection of the welder.

4.9 Auxiliary power. To assure compliance with 3.4.5 the auxiliary power receptacles shall be connected to a resistance load bank and load applied from 0 to 35 amperes. The regulation and output shall be as specified in 3.4.5.

4.10 Welding generator (Drawing number 11021792).

4.10.1 Welding output rating range. To assure compliance with 3.4.6.1 in the CC mode, with the welding generator connected to a resistance load bank, data shall indicate the unit capable of producing a minimum output not greater than 40 amperes at 21.6 volts and a maximum output of not less than 437.5 amperes at 37.5 volts (standard for NEMA rating of 350 amperes). There shall be no evidence of lack of overlap of welding current as plotted on the NEMA voltage line at any setting. Further:

(a) In the CV mode in the multi-wire output system, data shall indicate the ability to obtain a maximum output no less than 350 amperes at 34 volts and a minimum output no greater than 25 amperes at 21 volts.
(b) In the CV mode in the micro-wire output system data shall indicate the ability to obtain a maximum output no less than 250 amperes at 30 volts and a minimum output no greater than 20 amperes at 20.8 volts.

Temperature tests shall be conducted in accordance with NEMA EW-1 using the thermometer or resistance method as applicable. Temperature rise shall be within the

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limits as specified in NEMA EW-1.

4.10.2 Circuit voltage. To assure compliance with 3.4.6.2 the data shall indicate the maximum open circuit voltage shall not exceed 100 volts. In the CC mode the volt-ampere data shall indicate that the steady state short circuit current shall not be greater than 200 percent nor less than 110 percent of the normal current for any particular setting. With the machine operating in the CC mode and no load applied an oscilloscope shall be connected to the output terminals to visually observe the ripple voltage of the welder output. The ripple voltage shall not exceed 10 percent of the open circuit voltage.

4.10.3 Welding performance. To assure compliance with 3.4.6.3 and NEMA EW-1 the welding performance tests shall be performed by a trained operator. Welds on test plates shall be visually examined for acceptable quality. Failure of the welding machine to perform any of the tests in 4. 10.3.1 and 4.10.3.2 shall be cause for rejection of the welder.

4.10.3.1 Welding tests in the CC mode. Welding tests in the CC mode shall be as follows:

(a) Using the welding leads specified in paragraph 3.3.10 (50 ft (15.24 m) lengths), welding operations shall be performed using the SMAW process with 5/32 inch (3.97 mm) diameter E-6010 electrodes on mild steel plates, 2 pieces approximately 2 inches (50.8 mm) by 10 inches (254 mm) by 1/4 inch (6.35 mm) in a horizontal fillet position, tee joint. Welds shall be 10 (254 mm) long.

(b) Using the above welding leads, arc gouging torch, air supply and appropriate accessories (contractor supplied), cutting and gouging shall be performed (AAC process) using 5/32 inch ((3.97 mm) diameter copper coated carbon electrodes

10 inches (254 mm) by 1/2 inch (12.7 mm). Cut one piece across the 4 inch (101.6 mm) width and gouge the other 10 inch (254 mm) length.

(c) Using a gas shielding tungsten electrode welding set (TIC torch), hose, cables and regulator supplied by the contractor, with Argon gas, 1/16 inch (1.59 mm) diameter 2 percent thoriated tungsten, perform a horizontal fillet lap joint weld using 2 pieces of 2 inch (50.8 mm) by 10 inch (254 mm) by 16 gauge stainless steel. Using 1/8 inch (3.18 mm) diameter 2 percent thoriated tungsten, perform a horizontal fillet joint weld using 2 pieces of 2 inch (50.8 mm) by 10 inch (254 mm) by 1/8 inch (3.18 mm) stainless steel. Welds shall be 10 inches (254 mm) long.

4.10.3.2 Welding Tests in the CV mode. Welding tests in the CV mode shall be as follows:

(a) Perform a horizontal fillet tee joint weld using .035 inch (.889 mm) diameter AWS A5.18 classification ER-70S-3 wire on test Plates of 2 Pieces of 2 inch (50.8 mm) by 10 inch (254 mm) by 1/8 inch (3.18 mm) mild steel. The contractor shall supply the gas shielded consumable electrode welding set, gas hose, regulator flowmeter, cylinder of C02 and any other necessary accessories required to perform this test. Welds shall be 10 inches (254 mm) long.

(b) Using the welding set and accessories specified above, perform a horizontal fillet tee joint weld using l/16 inch (1.59 mm) diameter, flux cored wire on test plates of 2 pieces of 4 inch (101.6 mm) by 10 inch (254 mm) by 3/8 inch (9.53 mm) mild steel. Welds shall be 10 inches (254 mm) long.

(c) Using the welding set and accessories specified above, connected to a cylinder of Argon gas, perform a horizontal fillet lap joint weld using .035 inch (.889 mm) diameter aluminum welding wire ER-4043 on test plates consisting of 2 pieces of 2 inch (50.8 mm) by 10 inch (254 mm) by 1/8 inch

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(3.18 mm), 1,100 series aluminum. Welds shall be 10 inches (254 mm) long.

(d) Using the welding set and accessories specified above, connected to a cylinder of Argon gas, perform a fillet lap joint weld using 3/64 inch ((1.19 mm) diameter aluminum welding wire ER-4043 on test plates consisting of 2 pieces of 2 inch (50.8 mm) by 10 inch (254 mm) by 1/4 inch (6.35 mm) 1,100 series aluminum. Welds shall be 10 inches (254 mm) long.

4.11 Rectifiers (Drawing numbers 11022586 and 11022587). To assure compliance with 3.5 the temperature rise of the rectifiers, with the machine operating at full rated load, shall not exceed the manufacturer's recommendations. The temperature of the rectifiers shall be measured by the thermocouple method (see 6.3) after the machine has operated for one hour at rated load. This test may be run in conjunction with the generator temperature test (see 4.10.1).

4.12 Control panel (Drawing number 11021513). The control panel shall be visually inspected for compliance with the drawing and 3.6.

4.13 Dielectric strength. To assure compliance with 3.7, a high potential strength test shall be conducted in accordance with NEMA EW-1.

4.14 Stability. To assure compliance with 3.8 the unit shall be operated, in each of four planes 15 degrees from the horizontal, in the CC mode at rated load for a period of four hours, each position. All engine and generator controls shall be exercised at least once each hour. All components shall function normally and shall show no evidence of low lubricating oil pressure or excessive coolant temperature.

4.15 Operating speed. To assure compliance with 3.9 in the CC mode with the unit connected to a resistance load bank, the auxiliary power output connected to a resistance load bank and with the air compressor operating, the unit shall operate from O to rated load. Speed regulation shall be as specified in paragraph 3.9.

4.16 Antifreeze. The liquid in the engine coolant system shall be tested, by means of a specific gravity test, to ascertain compliance with 3.10.

4.17 Fuel tank. The fuel tank shall be visually inspected to ascertain that it is drained prior to shipment (see 3.11).

4.18 Workmanship. Workmanship shall be visually inspected for compliance with 3.12.

4.19 Packaging inspection. The welding machine shall be inspected before and after packaging to determine compliance with the preservation, packaging, packing, and marking requirements specified in section 5 and the contract (see 6.2.1).

5 PACKAGING

5.1 Cleaning, drying and preservation application (methods of). Cleaning, drying and preservative types and methods shall be in accordance with applicability and guidelines of MIL-P-116.

5.2 Level A or B preservation and packing. Level .4 preservation requirements and with level A or B packing requirements of MIL-W-45562, as specified in the contract. Items not covered under MIL-W-45562 shall be preserved, and packed as specified in the following paragraphs or level A or B requirements of section 5 of the applicable commodity specification,

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5.3 Preservation Preservation shall be level A or industrial as specified (see 6.2.1).

5.3.1 Compressors. Compressors and components shall be preserved level A and packed in accordance with level A or B requirements of MIL-C-3600, as specified in the contract.

5.3.2 Wire feeders. Wire feeders shall be preserved and packed in accordance with method IIB of MIL-P-116 and shall also be cushioned to restrict movement and prevent damage.

5.3.3 Fuel tank. Interior surfaces of the fuel system shall be completetely fogged (tank and filters (s)) with type P-10, grade 30, preservative oil.

5.3.4 Miscellaneous supplies and equipment. Items, equipment and components, not specifically covered by the preservation requirements of this document, shall be preserved as specified for supplies and equipment of similar design and construction. Other applicable packaging documents to be used for related items when applicable, shall be MIL-T-45542 (Sets, kits & outfits), PP?-P-40 (Hand tools), MIL-W-45562 (Welding equipment), and MIL-E-16298 or MIL-E-17555 (Electric or Electronic).

5.3.5 Preservation of lubrication systems. All lubrication systems shall contain supplier's luricants. However, in the event any areas for which the supplier's lubricant cannot be left in the machine (e.g. large areas which cannot be covered and sealed), applicable preservation paragraphs or preservation documents referenced in MIL-W-45562 shall be used. All vent openings in the lubrication systems shall be temporarily capped or plugged to prevent loss of lubricants during shipment. Tags, warning that "ALL TEMPORARY CAPS OR PLUGS MUST BE REMOVED BEFORE START-UP OR USE OF THE ITEM", shall be attached to all temporary caps or plugs intended to be vented to the atmosphere. These warnings shall also be included in the instruction manual. In the event a lubricant resevoir cannot be covered and sealed, the appropriate lubricant shall be provided in a sealed container and shipped with each machine. A WARNING TAG shall be placed in a conspicuous place near the filling area, WARNING THE USER TO INSTALL THE FURNISHED LUBRICANT BEFORE OPERATION OF THE EQUIPMENT. The operators manual shall also reflect the above warning and procedure.

5.3.6 Storage container(s). The parts, components, accessories or assemblies preserved and provided unit protection, as specified, shall be consolidated and placed in storage containers. Arrangements of contents, cushioning, blocking, bracing and anchoring shall be in accordance with MIL-STD-1186 to control movement and prevent damage to the items within. The storage containers themselves shall be packaged in fiberboard boxes conforming to PPP-B-636, to prevent marring of outside surface of the storage container. Hinges and latches shall be cushioned to prevent damage to the fiberboard box from within. However, if storage containers are not specified, any parts, components, accessories or assemblies, preserved and provided unit protection, as specified shall be boxed in fiberboard boxes conforming to PPP-B-636.

5.3.7 Industrial. Industrial preservation shall be as specified in ASTM D 3951.

5.4 Packing. Packing shall be level A, B, or industrial as specified (see 6.2.1).

5.4.1 Level A and B packing. The welder, along with all storage containers packaged and packed as secified, shall be packed in shipping containers conforming to MIL-C-104. The selection of a container shall be governed by the size and weight limitations of the container specification. Closure and strapping shall be in accordance with container specification except the strapping shall conform to

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QQ-S-781, type I of IV, finish A. Skid base of shipping container shall have forklift capabilities. Storage container boxes shall be strapped to base of shipping container to prevent movement during shipping and storage so as not to damage the welder. Edges of the storage container boxes shall be protected underneath the strapping. Arrangements of contents, cushioning, anchoring, blocking, and bracing shall be in accordance with MIL-STD-1186.

5.4.2 Industrial packing. Industrial packing shall be in accordance with ASTM D3951 .

5.5 Marking. Marking shall be military or industrial, as specified (see 6.2.1).

5.5.1 Military. When level A preservation and level A or B packing is specified, marking shall be in accordance with MIL-STD-129. Special markings required shall be as specified in the contract.

5.5.2 Industrial. Industrial marking shall be in accordance with ASTM D 3951 l

6 NOTES

6.1 Intended use. The machines covered by this specification are intended to provide a DC power source for welding where it is not practical to use electric power or it is not available, where portability is desired in shops, field maintenance and construction, and to provide auxiliary electric power as desired in connection with the welding operation.

6.2 Ordering data.

6.2.1 Acquisition data. Procurement documents should specify the following:

(a) Title, number and date of the document.
(b) First Article, when required (see 3.1)
(c) Mounted air compressor kit assembly when required (see 3.2).
(d) System of measurement required (see 3.2.2).
(e) First Article Inspection, when required (see 4.2.1).
(f) Packaging inspection, if different (see 4.19).
(g) Level of preservation and packaging (see 5.3 and 5.4).
(h) Marking, military or industrial (see 5.5)

6.2.2 Contract data requirements. Required technical data such as operators's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DE Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.3 Thermocouple method. To measure the rectifier temperature by the thermocouple method , a small hole is drilled in the case of the rectifier just deep enough to insert the ball of an iron constantan thermocouple. The thermocouple is held in place with epoxy. The opposite end of the thermocouple is connected to the appropriate readout equipment. The temperature measured by this method is the case temperature of the rectifier which can be converted to the rectifier junction temperature if necessary.

6.4 Subject term (key word) listing.

Antifreeze
Brazing
Castings

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Cleaning
Fastening Devices
Forgings
Fungus Resistance
Generator
Interchangeability
Measurement System
Oil, Lubricating
Painting
Rectifier
Safety Devices
Soldering
Threads
Transportability
Treating
Welding
Workmanship

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - AL

Preparing activity:
Army - AL

Project No. 3431-A174

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording.

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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